

Sampling:

$$x_0 \sim \pi_{old}^{rm}$$

x_t : M M M

Predict
↓

x_0 : London is a nice ...

Re-mask
↑

$$\{x_0^g\}_{g=1}^G$$



$$\pi_{old} = \pi_{\theta}$$

Policy Optimization

Consistent Forward

x_0

Forward
↓

x_t

Denoising Probs

$$\{\log p(x_0^n | x_t)\}_{k=1}^K$$
$$p : p_{\theta}, p_{old}, p_{ref}$$

Likelihood Ratio

$$\frac{p_{\theta}(x_0)}{p_{old}(x_0)} \approx \frac{\exp L_{\theta}(x_0)}{\exp L_{old}(x_0)}$$

ELBO-based RL

*TIM bias, Likelihood gap,
degraded performance...*

Guided Denoiser Self-Distillation

~~TIM bias~~, LL-free, off-policy